Yousef Elbaz

Machine Learning Engineer | AI Researcher

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LinkedIn | GitHub | Kaggle

Professional Summary

Machine Learning Engineer and AI Researcher with 3+ years of combined industry and academic research experience. Builds and deploys end-to-end AI and data systems spanning NLP, LLMs/RAG, computer vision, IoT analytics, text-to-SQL business intelligence, and recommender systems, with hands-on experience in MLOps and cloud deployment.

Technical Skills

Python | SQL | TensorFlow | Keras | PyTorch | Scikit-learn | HuggingFace Transformers | LangChain | LangGraph | FAISS | Docker | FastAPI | AWS EC2 | Google Cloud Platform | Power BI | Oracle EBS

Professional Experience

Teaching Assistant March 2025 - Present

Zewail City of Science and Technology

- Deliver lab sessions and assist in lecturing undergraduate courses in machine learning, data science, and applied AI.
- Support curriculum development and design practical assignments aligned with current industry standards.
- Mentor students on research methodologies, ML project design, and academic integrity.

Data Scientist January 2024 - March 2025

Egyswiss Holding Company

- Developed an end-to-end data pipeline extracting and processing business-critical data from Microsoft Dynamics 365 ERP via Python-based API integrations.
- Used SQL to clean, transform, and structure raw ERP data, ensuring consistency and analytical readiness.
- Built interactive Power BI dashboards visualizing KPIs for real-time decisions.

Data Scientist 2024 - 2026

Salab

- Designed and built a fully automated text-to-SQL analytics dashboard against Oracle E-Business Suite ERP.
- Automated recurring daily and weekly KPI reporting and feedback cycles for leadership.
- Consolidated cross-module ERP data into a unified reporting layer, reducing manual reporting effort.

Research & Projects

Research Intelligence Operating System (RIOS) 2025

Python, LangGraph, Chainlit, LLM APIs, arXiv API

- Engineered an AI-driven research platform for literature retrieval, structured planning, sandboxed code execution, validation, self-correction, and report generation.
- Integrated the arXiv API with caching, retry, backoff, and timeout handling.

Medical RAG Assistant 2025

Python, FastAPI, LangChain, FAISS, Docker

- Built a privacy-focused RAG system indexing 16,000 MedQuAD question-answer pairs with fully local inference.
- Containerized the application with Docker Compose, PostgreSQL, and Redis.

Quantum-Enhanced Deepfake Detection 2024 - 2025

Quantum ML, PyTorch, Explainable AI

- Conducted original research integrating quantum circuits into classical neural network architectures for deepfake detection.
- Authored a research paper demonstrating measurable improvements in optimization and model efficiency.

Education

Pre-Master's Program (Completed) 2024 - 2025

Zewail City of Science and Technology

B.Sc. Computer and Information Sciences, GPA 3.89/4.0 2019 - 2023

Zagazig University - Computer Science

Certifications

- Google Data Analytics Professional Certificate - Coursera
- Career Track: Data Science - 365 Data Science

Awards & Achievements

- 2nd Place - Dell Envision the Future Competition, MENA Region, Africa & Turkey
- 3rd Place - EUI Climathon, Smart Transportation Theme, MENA Region & Africa

Research Interests

Quantum Machine Learning | Explainable AI | Large Language Models | Arabic NLP